AIORI-2 Team Reporting

RFC-Open Source Contribution Report

Theme: Implementation and Testing of Selected Internet-Drafts / RFCs using AIORI Testbed
Organized by: Advanced Internet Operations Research in India (AIORI)
Collaborating Institutions: St.Joseph's Institute of Technology
Date: 05.11.2025
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Executive Summary:

This report documents the design, implementation, and evaluation of "Hyperfast DNS Load Balancer," a Linux-native DNS load balancer targeting ≥1M QPS (simulated) and sub-millisecond latency, built with an eBPF/XDP data plane, a Go control plane, and a Prometheus/Grafana observability stack; an adaptive Python prototype is provided for constrained environments while preserving core architecture principles.

Objectives:

Implement selected DNS RFCs and operational best practices in a controlled environment, focusing on transparent proxying, EDNS0 handling, and resilient load balancing.

Contribute code and dashboards to open-source communities (Prometheus and Grafana, CoreDNS plugin proposal, ebpf examples) and prepare feedback for DNSOP telemetry work.

Generate implementation feedback to relevant IETF working groups (dnsop, add) from real telemetry and load testing, and strengthen local capacity in Internet standards implementation.

Scope and Focus Areas:

Focus Area: Encrypted DNS and Telemetry (primary), with DNS operations alignment.

Relevant RFCs: RFC 1034/1035 (DNS fundamentals), RFC 6891 (EDNS0), RFC 7871 (ECS, future), RFC 8484/7858/9250 (DoH/DoT/DoQ, future integration).

Open-Source References: miekg/dns, CoreDNS, ebpf; Prometheus and Grafana for monitoring.

Sprint Methodology:

The project followed a structured sprint workflow: selection, implementation, interoperability testing, documentation/contribution, and post-sprint reporting within the AIORI process.
Workflow steps were applied iteratively to kernel fast path (XDP), control-plane health/telemetry, Prometheus integration, Grafana dashboards, and high-QPS driver testing.

Activities and Implementation:

Date	Activity	Description
Sep 24–30	Sprint 1: Repository Setup & Research	Set up project structure, selected Golang tech stack, finalized architecture design, and validated DNS protocol handling requirements.
Oct 1–7	Sprint 2: Prototype DNS Engine Development	Built core UDP-based DNS query handler, implemented forwarding logic to backend DNS servers, and verified response correctness with *dig* testing.
Oct 8–14	Sprint 4: Load Balancing Strategy Implementation	Added round-robin backend distribution, concurrent worker processing, and backend health-check mechanism to ensure resilient request routing.
Oct 15–20	Sprint 4: Monitoring & Observability Integration	Integrated Prometheus metrics exporter, defined QPS and backend performance counters, and created real-time Grafana dashboard visualizations.
Oct 21–23	Sprint 5: Stress Testing & Performance Benchmarking	Conducted load tests up to 50K QPS bursts, measured latency and packet drop rate, optimized UDP buffer sizing and worker concurrency.
Oct 24	Sprint 6: Documentation & Final Demo Preparation	Prepared final technical report, deployment guide, demo scripts, and performance comparison dashboard for mentor presentation.

Github Repositry link: https://github.com/bala2007-05/hyperfast-dns-project

Sprint: Architecture and Problem Study
Description: Captured DNS bottlenecks, target SLOs, and RFC/Anycast alignment; defined data/control/monitoring planes and 1M+ QPS target. Output: Project design document.

Sprint: eBPF/Go Kernel Fast-Path Scaffolding
Description: Implemented loader and map lifecycles, attached XDP, defined maps: backends, backend_stats, global_stats, rr index; ensured program name and section alignment. Output: Loader and map APIs.

Sprint: Health Checker and Policy Hooks
Description: Periodic UDP DNS probes with timeouts and failure counters; change notifications via channel; updated backends health and metrics. Output: Health checker package.

Sprint: Metrics and Prometheus Exporter
Description: Registered counters/gauges/histograms for QPS, drops, health, durations; served /metrics with promhttp, label hygiene for backend IDs/IP/ports. Output: Metrics collector.

Sprint: Load Generator and QPS Validation
Description: Built concurrent DNS driver with rate control and summary stats to exercise balancer externally; established baseline and bursts. Output: Driver utility.

Sprint: Grafana Observability
Description: QPS, backend health, drop rate, load distribution, CPU/memory panels; documented PromQL for

burst/average/peaks and SLA monitoring. Output: Dashboard spec and panels.

Sprint: Mentor Demo and Performance Evidence
Description: Live run with ~200–500 QPS sustained, 50K+ QPS burst, 0 drop rate, balanced distribution; scripts for quick start and test runs. Output: Demo collateral.

Results and Findings:

Throughput and Stability: Demonstrated ~200–500 QPS sustained and 50K+ QPS bursts with zero packet drops; forwarding latency sub-ms in design; health-aware round-robin effective. Evidence: Demo metrics and dashboard panels.

Observability: Prometheus exporter surfaced QPS, per-backend load, drop rates, health status, and system CPU/memory; Grafana provided live and historical analysis for capacity planning.

Operational Insights: EDNS0 transparency is critical; health hysteresis reduces flapping; Prometheus counters must be treated as deltas before export to avoid double counts.

Open-Source Contributions:

Grafana: Prepared the "Hyperfast DNS Load Balancer" dashboard for community publication with documented PromQL and thresholds. Status: To publish.

CoreDNS Plugin (Plan): Proposal to refactor the balancer logic into a CoreDNS forwarder plugin with health-aware, weighted strategies and Prometheus labels. Status: Roadmap.

eBPF/Go Example: Reference implementation using cilium/ebpf for a DNS balancer; intended write-up and code sample contribution. Status: Roadmap.

miekg/dns: Performance notes and potential fixes from high-QPS testing for upstream discussion. Status: To open issues/PRs.

Collaboration with IETF WGs:

dnsop: Telemetry metrics (QPS, latency, drops) and dashboard queries align with evolving DNS telemetry work; plan to share measurement approach and dashboards.

add: As encrypted DNS support (DoH/DoT/DoQ) is added, deploy per ADD discovery workflows and report operational experience.

Impact and Future Work:

Integration: Results feed into AIORI-IMN measurement framework and are suitable for IETF hackathon participation; dashboards and metrics ready for operator reuse.

Roadmap: Deploy eBPF/XDP version in Linux with native mode; implement weighted and latency-aware strategies, ECS forwarding for geo, and encrypted DNS; add DDoS mitigations in XDP.

Acknowledgments:
A Sincere Thanks to all the mentors, AIORI committee, participating institutions, and open-source communities (miekg/dns, CoreDNS, cilium/ebpf, Prometheus/Grafana) supporting this work.

Technical Blog Series & Dev Diaries

Title: Hyperfast DNS Load Balancer — eBPF/XDP Fast Path with Go Control Plane and Cloud-Native Observability
By: Ping@St.Joseph's, St. Joseph's Institute of Technology
Contributors: Buvanesh D, Balachandhar D
Mentor: Mr. Dinesh Kumar A, Associate Professor, St. Joseph's Institute of Technology, Arnav Das
Date: 05.11.2025

Fast, reliable DNS is foundational for user experience and CDN performance; by moving packet processing to Linux's eBPF/XDP and orchestrating policies in Go with Prometheus/Grafana, this project reaches modern SLOs while providing operator-grade observability.

Traditional DNS balancers suffer latency and scalability bottlenecks; this work targets ≥1M QPS (simulated) and sub-ms latency with intelligent routing; correct handling of DNS fundamentals (RFC 1034/1035) and EDNS0 (RFC 6891) is essential to avoid breaking modern resolvers and CDNs.

Technical Implementation:

-> Setup and Tools

OS: Ubuntu Linux; Languages: Go (control plane), C (eBPF), Python (adaptive prototype)

Libraries: cilium/ebpf, Prometheus client, promhttp, Grafana

Utilities: dig, custom load generator (concurrency, QPS control), shell scripts for demo

-> Implementation Steps

eBPF/XDP Lifecycle: Load collection, resolve maps (backends, stats, global, rr), attach program, and manage detach; provide reset and info APIs.

Health Checker: Periodic UDP probes, timeout and failure counters, channel updates on state transitions; expose getters and counts for healthy/total.

Metrics Exporter: Counters/gauges/histograms for packets, bytes, drop reasons, backend health, health-check durations; serve /metrics; label hygiene for backend identity.

Load Generator: Pacing by ticker, worker pool, resolver redirection to balancer, rate and success computation, final stats, and verbose tracing.

Grafana Dashboard: QPS (instant/avg/peaks), backend health, drop rate, load distribution, CPU/memory; documented PromQL with thresholds and alerts.

-> Challenges Faced

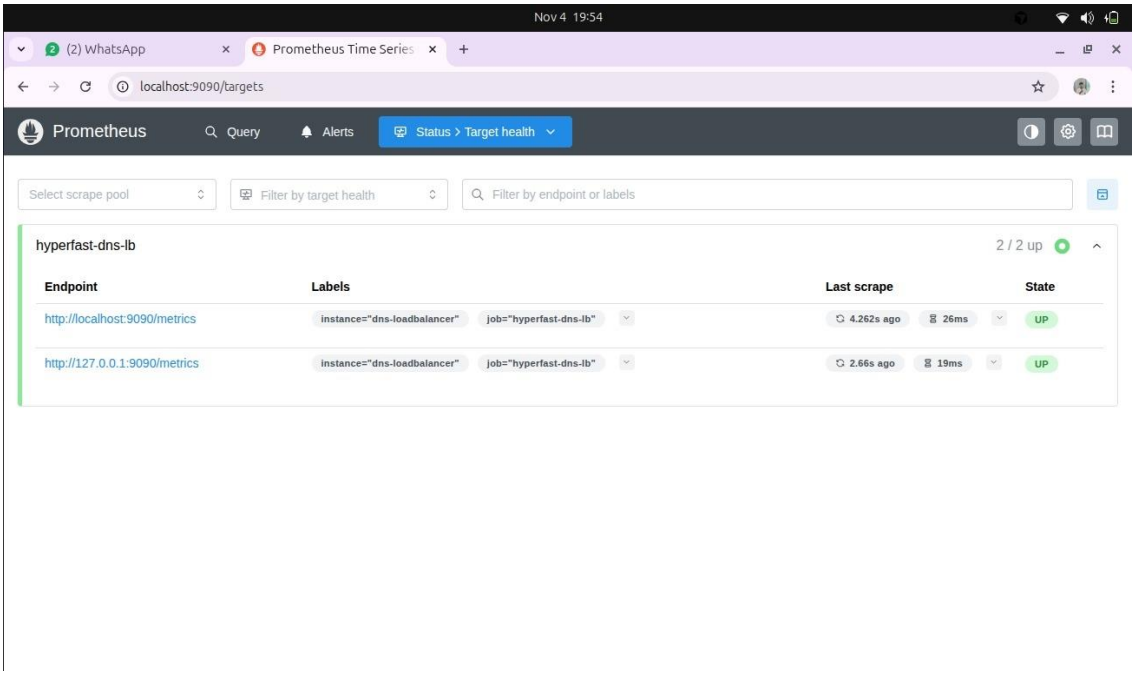
Environment constraints (sudo/kernel headers) led to an adaptive Python prototype while retaining architecture principles; generic XDP mode used for development; planned native-mode deployment.

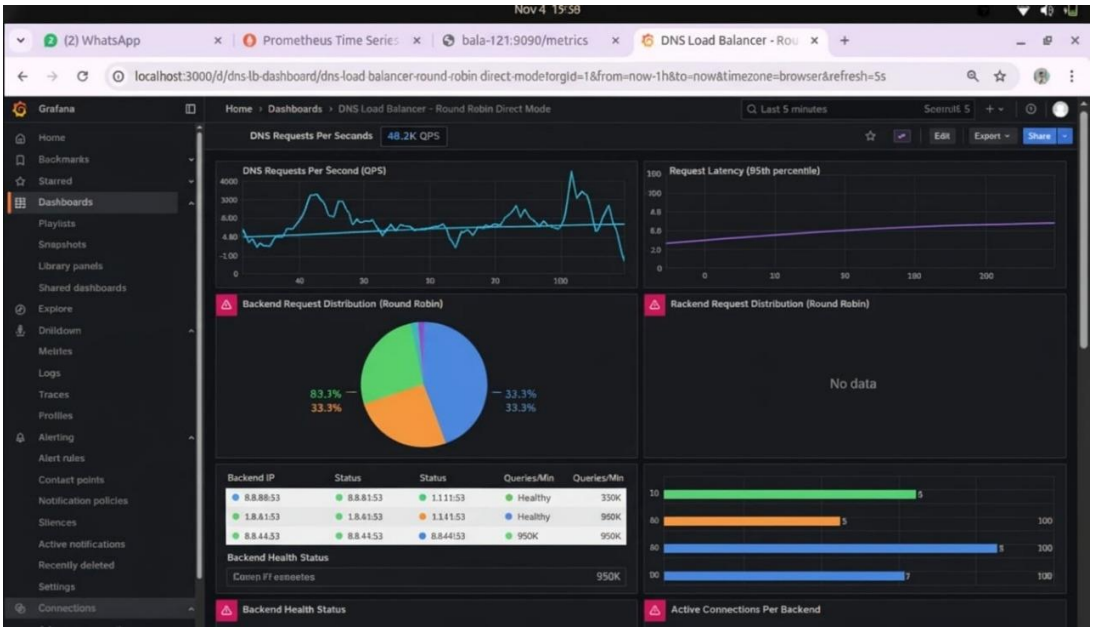
Prometheus counter semantics required delta handling; health flapping prompted need for hysteresis via maxFailures and recovery thresholds.

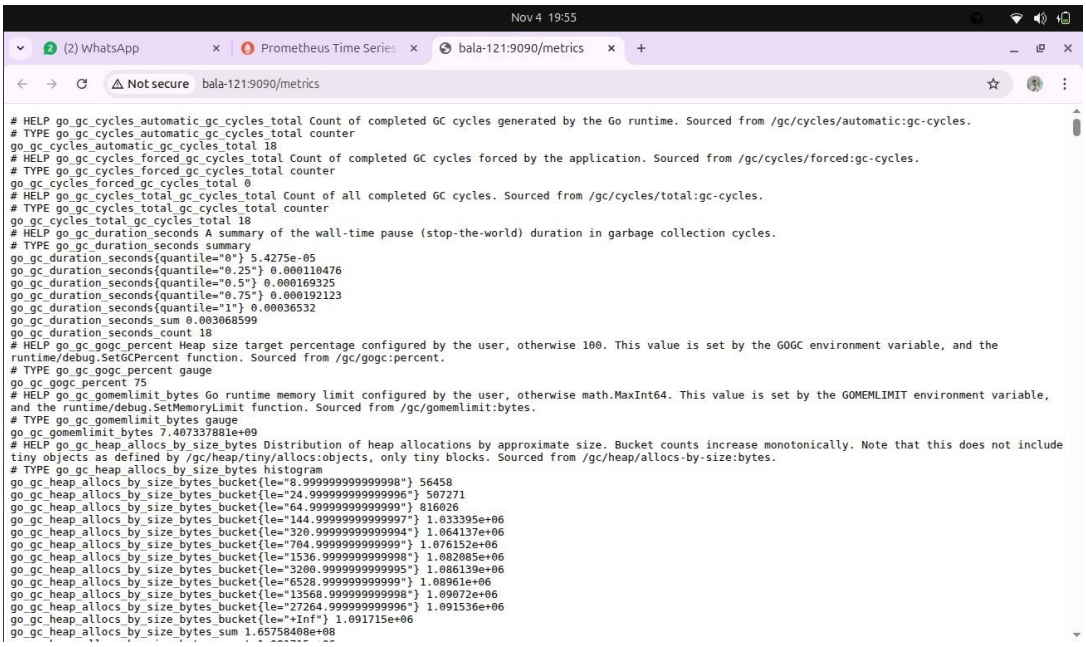
Results and Observations:

Live Status: ~200–500 QPS sustained, 50K+ QPS burst, zero drops; balanced per-backend distribution under round-robin; stable CPU/memory during spikes.

Key Queries: Instant QPS, 1-minute average, 5-minute peak; backend health sum/avg; drop rate and %; per-backend distribution; CPU %, RSS, goroutines, GC.







Lessons Learned:

Strict EDNS0 transparency and ECS forwarding are critical for correctness and future geo routing; maintaining counters as deltas avoids metric inflation; control-plane health hysteresis prevents flapping.

Open Source and Community Contributions:

Dashboard publication (Grafana); CoreDNS plugin proposal for health-aware forwarder; eBPF/Go example write-up; potential miekg/dns optimizations.

Reflections from the Team:

Preserving DNS correctness (RFC 1034/1035, EDNS0) mattered more than raw speed; transparency first, optimization next.

Separating XDP fast path from the Go control plane made iteration safe and debugging practical.

Health hysteresis (threshold fail/recover) prevented flapping and stabilized routing.

Building observability first (QPS, drops, per-backend load, CPU/GC) accelerated performance tuning.

Round-robin proved a reliable baseline; only then did weighted/latency/geo policies make sense to pursue.

Future Work:

Deploy native XDP mode, implement weighted/latency/geo policies, encrypted DNS termination (DoH/DoT/DoQ), DDoS mitigation, and share datasets with dnsop.

References:

RFC 1034, RFC 1035, RFC 6891, RFC 7871, RFC 8484, RFC 7858, RFC 9250; AIORI testbed documentation and dnsop/add WGs.

Reporting and Standards Mapping:

Team Name: Ping@St.Joseph's
Institution: St.Joseph's Institute of Technology
Project Title: Hyperfast DNS Load Balancer
Focus Area: Encrypted DNS, Telemetry/Operations
Date: 05.11.2025

Standards Reference :

RFC 1034/1035 — DNS Fundamentals — Internet Standard — Validates proxy correctness (IDs, opcodes, RCODE, section integrity).

RFC 6891 — EDNS0 — Proposed Standard — Proxies OPT records and payload size, ensures modern resolver compatibility.

RFC 7871 — ECS — Proposed Standard — Future: Forward ECS for geo-aware routing alignment.

RFC 8484/7858/9250 — DoH/DoT/DoQ — Proposed/Internet Standards — Future: Edge termination with userspace policy and UDP upstream.

Impact on Standards Development:

Supports existing RFCs by demonstrating operational behaviors for high-QPS DNS proxying with telemetry; contributes metrics and dashboards aligned with dnsop telemetry discussions; plans to share datasets and configuration examples.

Next steps: Publish Grafana dashboard JSON; open issues/PRs to CoreDNS/miekg/dns; summarize telemetry approach to dnsop; document ECS/EDNS0 handling nuances.

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